

LING 510: Week 2 review

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Office hours: W 11–1PM, N421H | [Zoom](#)



Plan for the day: Review concepts, practice, practice, your questions

1 Variety of inferences

1.1 Entailments

Truth conditions allow us to neatly capture logical relationships between sentences.

- (1) a. Laa-Laa likes singing and dancing.
 b. Laa-Laa likes singing.

Observation: When (1-a) is true, (1-b) automatically has to be true.

Entailment: For sentences S_1 and S_2 , *whenever* S_1 is true, S_2 is also true, we say that S_1 entails S_2 .

- This means that the truth conditions of S_1 are also always the truth conditions of S_2 !

Note the emphasis on ‘whenever’. That means, if you find S_2 to be only sometimes (or never) true while keeping S_1 true, there is no entailment between S_1 and S_2 .

Corollary: Suppose S_1 entails S_2 , then whenever S_2 is not true then S_1 is also not true.

Putting the definition and the corollary together we get a diagnostic to determine entailment relationships. we ask what happens when S_1 is true *and* S_2 is not true?

Contradiction test for entailment

S_1 entails S_2 , if and only if the sentence ‘ S_1 *and/but it is not the case that* S_2 ’ is a contradiction.

Demonstration:

- (2) #*Laa-Laa likes singing and dancing* but it is not the case that *Laa-Laa likes singing*.

There is no situation which would make (2) true. Therefore, it is a contradiction.¹

Conclusion: Laa-Laa likes singing and dancing $\Rightarrow$ Laa-Laa likes singing.

You will have noticed that sometimes two sentences entail each other i.e., they have identical truth conditions – Every situation that makes S_1 true, makes S_2 true and vice versa.

Also note that if two sentences mean (intuitively) the same thing then they entail each other. Verify! (same meaning $\rightarrow$ same truth conditions $\rightarrow$ entailment)

- (3) John is a bachelor. $\Leftrightarrow$ John is an unmarried man.

¹Contradictions are always false.

Logical equivalence

If S_1 entails S_2 and S_2 entails S_1 then we say that S_1 and S_2 are logically equivalent.

Corollary: If there is a situation s in which S_1 is true but S_2 is false (or simply that they have different truth values) then they are not logically equivalent.

Now consider the entailment patterns for this data. How does it change?

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| (4) a. John read a Russian book.
b. John read a book. | (7) a. Possibly, John read a Russian book.
b. John read a book. |
| (5) a. Allegedly, John read a Russian book.
b. John read a book. | (8) a. John rarely reads Russian books.
b. John rarely reads books. |
| (6) a. John often reads Russian books.
b. John often reads books. | (9) a. John read every Russian book.
b. John read every book. |

1.2 Presuppositions

Sometimes it happens that $S_1 \Rightarrow S_2$ but being able to tell whether S_1 is true or not depends on whether S_2 is true in the first place at all.

Consider the following two minimal pairs. The entailments hold.

- (10) John is French. $\Rightarrow$ John is European.
(#*John is French but its not the case that John is European.*)
- (11) John's sister is French. $\Rightarrow$ John has a sister.
(#*John's sister is French but its not the case that John has a sister*)

Now observe the following contrast. In some sense, even if you deny John having a sister, while denying, it feels weird to say that A's assertion is not true at the same time.

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| (12) A: John is French.
B: That's not true! John isn't (even) European. | (13) A: John's sister is French.
B: #That's not true! John doesn't have a sister. |
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In some sense the entailment in (11) is special because of the observation in (13). **You need John has a sister to be true to even begin to test whether its true that John's sister is French.**

Presupposition

Sentence S_1 presupposes that $p \Leftrightarrow S_1$ is true or false only if p is true.

S_1 can only have a truth value if p true.

Note that negation disrupts entailment in (14) but not in (15).

- (14) John is not French $\nRightarrow$ John is European.
(*John is not french but its not the case that John is European.*)
- (15) John's sister is not French. $\Rightarrow$ John has a sister.
(#*John's sister is not French but its not the case that John has a sister*)

Now consider that entailments can also be disrupted in other contexts but our pattern in (14) and (15) still survives.

Entailments disrupted:

- (16) a. Is John French? $\nRightarrow$
 b. John is European.
- (17) a. Its possible that John is French. $\nRightarrow$
 b. John is European.

Entailments survive:

- (18) a. Is John's sister French? $\Rightarrow$
 b. John has a sister.
- (19) a. Its possible that John's sister is French $\Rightarrow$
 b. John has a sister.

Entailment vs. Presupposition

p entails q $\nRightarrow$ If q is false then p is false
 p presupposes q $\nRightarrow$ If q is false then p is not defined.

Our observation that the presuppositions survive entailment disrupting environments gives a diagnostic test for presuppositions!

Diagnostic test for presuppositions

If S_1 is a sentence and we want to test if p is presupposed by S_1 , then

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|---|----------------------|
| 1. Does the sentence <i>Its not the case that</i> S_1 entail p? | negation test |
| 2. Does the question <i>Is it the case that</i> S_1 ? entail p? | question test |

If the answer to both is yes, then S_1 presupposes p

Demonstration:

- (20) John regrets that Mary fell.

Does (20) presuppose that *Mary fell*? yes!

- (21) **negation test** ✓

- a. Its not the case that John regrets that Mary fell $\Rightarrow$ Mary fell
 b. #*John doesn't regret that Mary fell but Mary did not fall.*

- (22) **question test** ✓

- a. Is it the case that John regrets that Mary fell? $\Rightarrow$ Mary fell
 b. #*Mary did not fall but does John regret that Mary fell?*

1.3 Implicatures

Consider the following dialogue where we learn something more from B's response.

- (23) A: Do you swim?
 B: I used to be a swimmer.
 $\nRightarrow$ *B doesn't swim.*

I am not a swimmer now is not an entailment (which means its not presupposed as well).

(24) *I used to be a swimmer but it's not the case that I don't swim.* (no contradiction)

The how do we get the inference? via reasoning.

(25) *B said they were a swimmer at some point before now. If B was still swimming, B would've said so which means B is not a swimmer anymore. Therefore, B doesn't swim.*

Now consider the following pair:

(26) *I used to be a swimmer. In fact, I still swim regularly.*
 $\rightarrow$ *I am not a swimmer now.* (Inference canceled)

(27) *I used to be a swimmer. In fact, I haven't swum in twenty years.*
 $\rightarrow$ *I am not a swimmer now.* (Inference reinforced)

Inferences we can reasonably draw from utterances in conversations are called **implicatures**. They can be **cancelled** without contradiction or undefinedness. the ' $\rightarrow$ ' indicates implicature.

Question: Identify (using tests) whether the following inferences are entailments, presuppositions, or implicatures.

- (28) a. John tried to eat Natto. $\sim\rightarrow$ *John didn't eat Natto.*
b. Mary read *Diaspora* or *Exordia*. $\sim\rightarrow$ *Mary didn't read both books.*
c. The building's bathroom is on the first floor or there is no bathroom
 $\sim\rightarrow$ The building has a bathroom
d. Alfonso discovered that Kim drives an expensive car.
 $\sim\rightarrow$ *Kim knows how to drive.*
- (29) a. It is Harry who broke the vase.
 $\sim\rightarrow$ *Someone broke the vase.*
b. Either there is no bathroom in the building or the building's bathroom is on the first floor
 $\sim\rightarrow$ The building has a bathroom
c. Bill heard that Steve stopped smoking.
 $\sim\rightarrow_1$ *Steve stopped smoking.*
 $\sim\rightarrow_2$ *Steve used to smoke.*
d. Some students enjoy Semantics. $\sim\rightarrow$ *Not every student enjoys Semantics.*

Things to remember:

1. Entailments cannot be cancelled without contradiction.
2. presuppositions survive entailment disruption.
3. Implicatures can be cancelled.

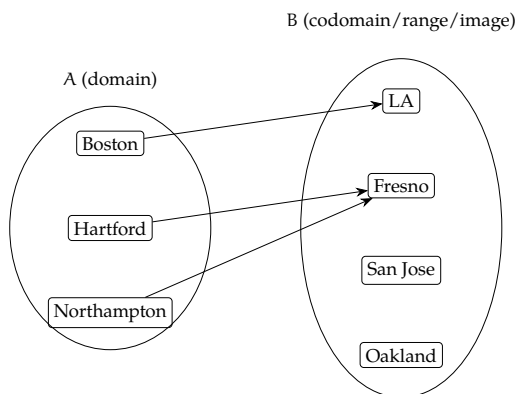
2 Mathematical preliminaries

The tacit assumption for us is that even though language and its rules might seem arbitrary, it has a logical structure underlyingly. Our job is to uncover and describe this structure as accurately as possible. We have made good on the first step – 1. Compositionality 2. Syntax 3. Meaning as truth

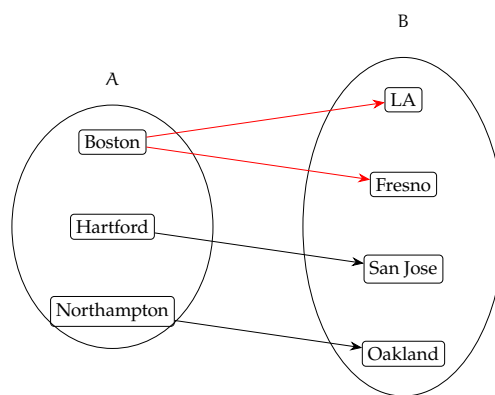
conditions. Now we will develop our formal machinery and have a toolkit to get to task.....by the power of Math. Turns out, most everything when it comes to meaning can be modeled in terms of functions from set of things to set of things. So we need to know what functions are.

2.1 Functions

A function is like a map. It tells you what location you will land on if you start from a given location. For the map to be 'good', it should be unambiguous i.e., it should tell you exactly where you will land given a starting location. In other words, **each starting location must have a unique (i.e., one and only one) landing location**. This is shown below, where **1a** describes a 'well-defined' function and **1b**, which is not a function.



(a) This **is** a function: Boston, Hartford, Northampton each send out exactly one arrow. (It's fine that San Jose, Oakland receive nothing, and that Hartford, Northampton both point to Fresno.)



(b) This is **not** a function: Boston sends out **two** arrows (to LA *and* Fresno), so it's ambiguous where you would "land" from Boston.

If I wanted to send the map to someone else, how would I do it? I can think of a couple of ways of doing this: 1. Listing the pairs of starting and landing sits 2. A table 3. Something more compact. They all represent the same phenomenon.

$$1. f = \{ \langle \text{Boston}, \text{LA} \rangle, \langle \text{Hartford}, \text{Fresno} \rangle, \langle \text{Northampton}, \text{Fresno} \rangle \}$$

$$2. f = \left[\begin{array}{l} \text{Boston} \rightarrow \text{LA} \\ \text{Hartford} \rightarrow \text{Fresno} \\ \text{Northampton} \rightarrow \text{Fresno} \end{array} \right]$$

$$3. \text{ For } A = \{\text{Boston}, \text{Hartford}, \text{Northampton}\}, B = \{\text{LA}, \text{Fresno}, \text{San Jose}, \text{Oakland}\}, \\ f : A \rightarrow B \text{ such that } f(\text{Boston}) = \text{LA}, f(\text{Hartford}) = f(\text{Northampton}) = \text{Fresno} \quad (\text{better!})$$

Let's do better. I can actually describe the domains and codomains in a way that covers what is common to all the members in respective *sets* i.e., I can describe them by way of a property.

$$(30) \quad \text{Let } A = \{x : x \text{ is a town in Massachusetts}\}, B = \{y : y \text{ is a town in California}\}, \\ f : A \rightarrow B \text{ such that } f(\text{Hartford}) = f(\text{Northampton}) = \text{Fresno}$$

Takeaway: A function from a set A to set B is a **set** of *ordered pairs* where the first member (from A) in each ordered pair has a unique corresponding member from set B. A function is a well-defined **relation** on the set $A \times B$ (the set of all ordered pairs from A to B)

We now need a way to talk about how things relate to each other and whether something is in a set or not and what sort of relationships hold between different sets. Intuitively, if you have two sets, you can wonder about 1. what is in them? 2. what is *not* in them? 3. what is common between them? 4. what do you get if you put them together? 5. what is the difference between them? 6. what if a set has nothing? ...

1. Understand $a \in A$ to mean 'a is a member of A'. We write $a \notin A$ when it is not. You can either list all the members of A or you can be economical and smart. You can come up with a description that characterizes only the members of the set and nothing else. This is called **set abstraction** format. The following two represent the same things:

$$- B = \{2, 3, 5, 7, 11, 13, 17\}$$

$$- B = \{y : y < 18 \text{ and } y \text{ is a prime number}\} \quad (B \text{ is a set of primes less than } 18)$$

To tell whether some $a \in A$, check whether a satisfies the criterion for membership of A.

Demonstration: $19 \notin B$ since $19 > 18$, $16 \notin B$ since 16 is not a prime.

2. The things that two sets have in common also makes up a set. Understand $A \cap B$ to mean intersection of sets A and B.

$$A \cap B := \{z : z \in A \text{ and } z \in B\}$$

3. A set made out of members of both A and B is called the union. Understand $A \cup B$ to mean A union B.

$$A \cup B := \{y : y \in A \text{ or } y \in B\}$$

4. Things that are in A but not in B also make up a set. Understand $A - B$ to mean difference between A and B.

$$A - B = \{u : u \in A \text{ and } u \notin B\}$$

5. If a set has nothing, it is called the empty set $\emptyset$.

6. If every member of A is a member of B, We say $A \subseteq B$, A is a subset of B.

$$A \subseteq B \text{ iff } a \in A \Rightarrow a \in B$$

Questions:

1. Is 'cookie monster $\in \{x : x \text{ is blue}\}$ ' true?
2. What are the members of the set $\{x : \text{the sky is blue}\}$?
3. $\{x : x \text{ is blue}\} \cap \{x : x \text{ has four wheels}\} = ?$
4. Is $\{x : x \text{ is blue}\} \subseteq \{x : x \text{ is long}\}$?
5. $\{x : x \text{ is blue}\} \cap \{\{\text{cookie monster}\}\} = ?$
6. Does Cookie monster $\in \{y : \{y\} \cap \{x : \{x\} \subseteq \{x : x \text{ is a muppet}\} \cap \{x : x \text{ is blue}\}\} \neq \emptyset$?
7. Is it true that for $f: \{y : y \in \mathbb{N} \text{ and } y \leq 5\} \rightarrow \mathbb{N}$, for every $x \in \{y : y \in \mathbb{N} \text{ and } y \leq 5\}$, we have that $f(x) = 2x$
8. $f : \mathbb{N} \rightarrow \mathbb{N}$ we have that $f(x) = \sqrt{(x)}$, is this a function?
9. Write the following function $\{< 4, 2 >, < 6, 3 >, < 10, 5 >, < 14, 7 >, < 22, 11 >, < 26, 13 >, < 34, 17 >\}$ in the table format and set abstraction format.